1 Assignment 2 - Linear regression and ridge regression

1.1 Data and ordinary linear regression

We predicted motor UPDRS from 16 voice characteristics in the Parkinson's dataset. The data were randomly divided into 3525 training observations (60%) and 2350 test observations (40%), using `set.seed(12345)`. Both the predictors and response were standardized using the *training-set* means and standard deviations, and the same transformation was applied to the test set. Consequently, the models were fitted without an intercept.

The ordinary least squares (OLS) model had training MSE 0.8785 and test MSE 0.9354 (on the standardized response scale). 8 coefficients were significant at the 5% level: Jitter(Abs), Shimmer, Shimmer:APQ5, Shimmer:APQ11, NHR, HNR, DFA, PPE. These tests concern each coefficient conditional on the other predictors.

1.2 Likelihood, Ridge objective and degrees of freedom

We assumed independent Gaussian errors, with conditional response distribution and log-likelihood

$$Y_i \mid \mathbf{x}_i \sim N(\mathbf{x}_i^\top \theta, \sigma^2), \qquad \ell(\theta, \sigma) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top\theta)^2.$$

The implemented `Ridge` function evaluates the negative log-likelihood with an L2 penalty:

$$J(\theta, \sigma; \lambda) = -\ell(\theta, \sigma) + \lambda \sum_{j=1}^{16} \theta_j^2.$$

For each fixed λ, `RidgeOpt` minimizes this objective using `optim(method = "BFGS")`, jointly estimating the coefficients and σ. We optimized $\log\sigma$ instead of σ directly to enforce $\sigma > 0$.

Following Lecture 1d (slides 17 and 22), the `DF` function uses the trace of the Ridge smoother matrix to measure effective model complexity:

$$\mathrm{df}(\lambda) = \mathrm{tr}\left[X(X^\top X + \lambda I)^{-1} X^\top\right].$$

1.3 Comparison of penalty strengths

The fitted coefficients were used to predict responses by $\hat{\mathbf{y}} = X\hat{\theta}$. Table Table 1 compares the training and test MSE and the effective degrees of freedom for the three specified penalty strengths. The estimated error standard deviation is included for completeness.

Table 1: OLS and Ridge results; MSE values are on the standardized response scale.

Model	Training MSE	Test MSE	Effective df	Sigma
OLS	0.8785	0.9354	16.0000	0.9373
Ridge (lambda = 1)	0.8786	0.9350	13.8607	0.9374
Ridge (lambda = 100)	0.8844	0.9323	9.9249	0.9404
Ridge (lambda = 1000)	0.9211	0.9540	5.6439	0.9598

Among the specified penalties, $\lambda = 100$ produced the lowest test MSE (0.9323), compared with 0.9354 for OLS. The lecture-defined effective degrees of freedom decreased from 13.86 at $\lambda = 1$ to 5.64 at $\lambda = 1000$. Thus, stronger regularization reduced the model's effective flexibility, although all 16 coefficients remained in the model. The test MSE indicates which of these penalty strengths gave the best predictive performance on the held-out data.